

Daniel Costa Araújo

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RESEARCH INTERESTS

Statistical and array signal processing; parameter estimation in propagation channels; massive MIMO and MIMO-OFDM channel estimation; hybrid beamforming; localization and integrated sensing and communication (ISAC) for 5G/6G; machine learning for the physical layer; software-defined radio and microwave systems.

EDUCATION

- **Federal University of Ceará (UFC)** Fortaleza, Brazil
Postdoctoral Fellow in Teleinformatics Engineering – CNPq 2017 – 2018
- **Federal University of Ceará (UFC)** Fortaleza, Brazil
Ph.D. in Teleinformatics Engineering (CAPES/PDSE sandwich at Ericsson Research) Aug. 2012 – Sep. 2016
 - **Thesis:** Channel Estimation Techniques Applied to Massive MIMO Systems Using Sparsity and Statistical Approaches
 - **Advisors:** J. C. M. Mota (UFC); A. L. F. de Almeida (UFC, co-advisor); E. Karipidis (Ericsson)
- **Federal University of Ceará (UFC)** Fortaleza, Brazil
M.Sc. in Teleinformatics Engineering – CAPES Scholarship Aug. 2010 – Jun. 2012
- **University of Fortaleza (UNIFOR)** Fortaleza, Brazil
B.Sc. in Telecommunications Engineering Aug. 2005 – Jul. 2010

ACADEMIC POSITIONS

- **University of Brasília (UnB) – Faculdade do Gama** Brasília, DF, Brazil
Adjunct Professor December 2019 – Present
 - **Coordinator, B.Sc. in Electronic Engineering:** September 2023 – 2025
 - **Deputy Coordinator, B.Sc. in Electronic Engineering:** April 2020 – April 2023
 - **Vice-Coordinator:** LabTelecom – Shared Teaching and Research Laboratory in Telecommunications (LCEPT)
- **UFC – GTEL / Ericsson Research Program** Fortaleza, Brazil
Research Engineer (Wireless Telecommunications Research Group) Jan. 2012 – Dec. 2019
- **Ericsson Research AB** Kista, Sweden
Visiting Researcher – RAT (Radio Access Technology) Department May 2015 – Apr. 2016
- **Centro Universitário Christus / 7 de Setembro** Fortaleza, Brazil
Lecturer (Applied Math, Probability & Statistics, Digital Electronics, Linear Systems) Feb. 2017 – Jun. 2019

FUNDED RESEARCH PROJECTS (PRINCIPAL INVESTIGATOR)

- **FAPDF – 6G Localization & Sensing Platform:** Development testbed for angular profile estimation and waveform design for 6G ISAC (2022–2024)
- **FAPDF – Low-cost LEO satellite Internet platform:** Holographic and reflectarray antennas with ML-based beam steering for Starlink-class downlink (2022–2024)
- **CNPq – UNIESPAÇO 2022:** Distributed low-cost ground station with antenna arrays for LEO satellite downlink (2023–Present)
- **Industry partnerships:** Tecsys Industrial – SDR implementation of SIGFOX and LoRa protocols (2023–2024); SoC-FPGA digital TV system (2024)
- **SEMFOGO-DF:** AI/Computer Vision system for wildfire detection in the Brazilian Cerrado (2022–2024, with CBM-DF)

TEACHING (UNB)

- **Graduate (PPGEE):** Adaptive Array Signal Processing
- **Undergraduate (Electronic Eng.):** Communication Principles for Engineering; Signal Integrity in Circuit Design; Practical Electronic Circuits I

STUDENT SUPERVISION

- **M.Sc. Theses (principal advisor, completed):** 3 – L. A. A. Rocha (2021, near-field user localization), A. B. Bigonha (2023, cell-free MIMO clustering), G. B. Pinheiro (2025, adaptive array signal processing for LEO)
- **Ongoing:** 1 Ph.D. (A. C. P. Silva, coexistence studies in IMT systems) and 3 M.Sc. (B. C. Teixeira, UAV localization via distributed ISAC; H. O. Silva, lens-based DOA estimation and 77 GHz radar; K. Kuster, LEO orbit tracking with ground-station antenna arrays)
- **Ph.D. Thesis (co-advisor):** F. H. Costa Neto (UFC, 2020, ML-based radio resource management for 5G)
- **Undergraduate Research (IC) and Final Projects (TCC):** Numerous projects in sparse GNSS signal processing, cell-free MIMO beamforming, holographic antennas, Doppler estimation for LoRa-satellite links, and Butler matrices for directional beamforming
- **M.Sc. Examining Committees:** Member of 10+ M.Sc. defense committees in Electrical and Teleinformatics Engineering (UnB, UFC)

AWARDS

- **IET Communications Premium Award (2018):** Institution of Engineering and Technology, for the paper “Massive MIMO: Survey and Future Research Topics” (IET Communications, 2016)

PUBLICATIONS (SUMMARY)

- **Journal Articles:** 12 papers, including IEEE Wireless Communications Letters (3, 2026), IEEE Trans. Vehicular Technology, IEEE Access, IET Communications, Wireless Networks, Computer Communications, and Int. J. of Communication Systems
- **Conference Papers:** 25 papers in ICASSP, EUSIPCO, CAMSAP, LATINCOM, ITS, GLOBECOM Workshops, ISWCS, WCNPS, and SBrT
- **Patents:** 2 (WIPO international filing on RL-based uplink power control, WO 2021/223865; antenna beam control, US 2015/0264583)
- **Registered Software:** 1 – Computational Modeling for Indoor Localization in Industrial Environments (INPI BR 51 2026 001765-3, 2026)

EDITORIAL AND PROFESSIONAL SERVICE

- **Associate Editor:** IEEE Signal Processing Letters (2025 – present)
- **Associate Editor:** Signal Processing Open (Elsevier), 2026 – 2029
- **Reviewer:** IEEE Trans. Communications, IEEE Wireless Communications Letters, Signal Processing Open, Circuits, Systems, and Signal Processing

LANGUAGES

Portuguese (native) | English (fluent) | Spanish (intermediate reading) | French (basic).